

# DIDAR ALI

*Junior Data Scientist | Machine Learning | Data Analytics*

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## Professional Summary

Computer Systems Engineering graduate with hands-on experience in Python, SQL, data analytics, machine learning, data processing, computer vision, OCR, and production AI workflows. Experienced in developing and validating data pipelines, extracting structured information from unstructured data, investigating model-output discrepancies, and testing multi-stage AI systems.

Built end-to-end machine-learning solutions involving data cleaning, preprocessing, feature engineering, model training, validation, and visualization, including a Wav2Vec2 transformer-based Speech Emotion Recognition system achieving approximately 79% classification accuracy. Published research in transformer-based speech emotion recognition.

Seeking to apply strong analytical, programming, and machine-learning skills to data science, predictive modeling, forecasting, and data-driven decision-making.

## Technical Skills

**Data Science & Analytics:** Python, SQL, Pandas, NumPy, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Data Transformation, Feature Engineering, Statistical Analysis Fundamentals

**Machine Learning & AI:** Machine Learning, Transformer Models, Model Training, Model Evaluation, Classification, Computer Vision, OCR, Artificial Intelligence

**Data Visualization:** Matplotlib, Seaborn, Jupyter Notebook, Dashboard-Based Visualization, MS Excel

**Programming:** Python, SQL, C++, C, Bash

**Data & AI Pipelines:** Data Extraction, Data Validation, Pipeline Testing, Automated Data Processing, Structured & Unstructured Data Processing

**Engineering & Tools:** Git, GitHub, Linux, Docker, FastAPI, Jupyter Notebook

**Relevant Engineering Knowledge:** Digital Signal Processing, Signals & Systems, Digital Image Processing, Algorithms & Data Structures

## Professional Experience

**AI Engineer I at Xflow Research Inc. - Islamabad, Pakistan**

**Feb 2026 – May 2026**

- Worked with Python-based AI, computer-vision, OCR, identity-verification, and data-processing pipelines in a production environment.
- Developed OCR-based workflows to extract, structure, and validate information from documents.
- Investigated discrepancies between model outputs, extracted information, and downstream verification logic to identify sources of error.
- Tested and validated multi-stage AI/software pipelines before deployment.
- Integrated NFC-based e-passport chip-reading functionality for automated structured-data capture and verification.
- Applied systematic debugging, testing, validation, and analytical problem-solving to complex multi-component systems.

## Selected Data Science & Machine Learning Projects

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### 1. Speech Emotion Recognition System

Python | Wav2Vec2 | Transformers | Machine Learning | Signal Processing

- Built an end-to-end machine-learning pipeline covering data cleaning, preprocessing, augmentation, feature preparation, model training, validation, and evaluation.
- Fine-tuned a Wav2Vec2 transformer model on noisy speech data for multi-class emotion recognition.
- Achieved approximately 79% classification accuracy across emotion categories.
- Analyzed model predictions and classification errors to evaluate performance and improve reliability.
- Applied signal-processing and machine-learning techniques to unstructured audio data.
- Research from the project was published in VFAST Transactions on Software Engineering in 2025.

### 2. Identity Verification & Document Intelligence

Python | OCR | Computer Vision | Data Extraction | Validation

- Contributed to a multi-stage identity-verification pipeline involving document processing, OCR, structured data extraction, and automated validation.
- Developed workflows to transform unstructured passport and identity-document inputs into structured information.
- Investigated discrepancies between OCR outputs, extracted fields, and downstream verification logic.
- Integrated NFC-based e-passport chip reading for structured passport-data capture and verification.
- Tested individual components and end-to-end workflows to identify processing and validation errors.

### 3. AI Health Super App

Python | Machine Learning | Data Processing | Visualization

- Developed an AI-driven application combining data preprocessing, model prediction, evaluation, and visualization.
- Built data-processing workflows to transform raw inputs into model-ready datasets.
- Applied machine-learning techniques for prediction and analytical decision support.
- Presented model outputs through dashboard-style visualizations.

## Research Publication

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### Vocal Sentiments: Transformer-Based Speech Emotion Recognition

#### VFAST Transactions on Software Engineering - 2025 ([LINK](#))

Research focused on transformer-based machine learning for speech-emotion recognition, including data preprocessing, noisy audio processing, model fine-tuning, experimentation, and performance evaluation.

## Education

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### B.Sc. Computer Systems Engineering

*University of Engineering & Technology (UET), Peshawar*

2021 – 2025 | CGPA: 3.01 / 4.00

**Relevant Coursework:**

Artificial Intelligence, Data Structures & Algorithms, Digital Signal Processing, Signals & Systems, Digital Image Processing, Computer Organization & Architecture, Object-Oriented Programming, Control Systems, Embedded Systems

## Certifications

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**Google Data Analytics Professional Certificate** - Google / Coursera

**Google Cybersecurity Professional Certificate** - Google / Coursera

## Core Strengths

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- Analytical and quantitative problem solving
- Data cleaning and preprocessing
- Machine-learning experimentation
- AI/model output validation
- Structured and unstructured data processing
- Technical research
- Independent learning
- Technical documentation
- Debugging and root-cause analysis
- Cross-functional teamwork
- Clear written and verbal communication

## Additional Information

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**English:** Professional working proficiency

**Location:** Rawalpindi, Pakistan

**Relocation:** Open to relocation